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ADVANCED DIGITAL

DESIGN PROJECT

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Abstract

This project presents the design and verification of a smart traffic light controller with pedestrian support based on a Finite State Machine (FSM). The controller manages traffic flow at an intersection consisting of a main road, a side road, and a pedestrian crossing, where the main road is given default priority and remains green under normal conditions.

State transitions are triggered by the detection of vehicles on the side road or by pedestrian button requests. The system operates according to fixed timing constraints for each traffic light state and correctly supports queuing pedestrian requests when the side road is active.

The controller is implemented using Verilog HDL and verified through a self-checking testbench. Simulation results demonstrate correct functionality and confirm that the design meets all specified requirements.

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State Diagram

The finite state machine consists of six main states that represent different traffic light configurations. each state defines the status of the main road, side road, and pedestrian walk signal.

Table 1:Table of State Definition

State	State Name	Main Road Light	Side Road Light	Walk Signal	Definition
MG	Main Green	Green	Red	OFF	Default state. Main road has priority and remains green when there are no requests.
MY	Main Yellow	Yellow	Red	OFF	Transition state before stopping the main road.
AR	All Red	Red	Red	OFF	Safety state to clear the intersection before switching directions.
SG	Side Green	Red	Green	OFF	Allows vehicles on the side road to move when VS is detected.
SY	Side Yellow	Red	Yellow	OFF	Transition state before returning control to the main road.
PG	Pedestrian Walk	Red	Red	ON	Pedestrians are allowed to cross safely.

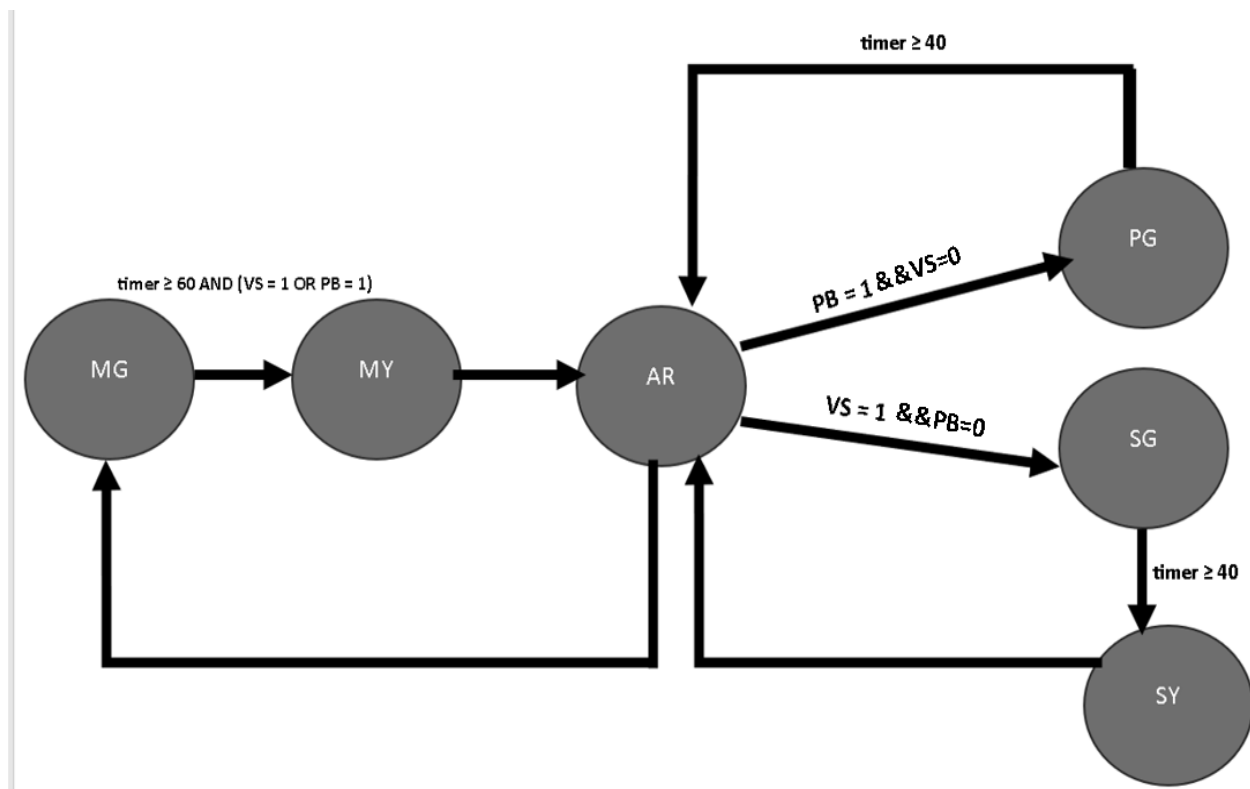


Figure 1:FSM State Diagram of the Traffic Light Controller

State Encoding

The traffic light controller is implemented using a Finite State Machine (FSM) in which binary encoding is used to represent the system states. The controller operates using six distinct states, each corresponding to a specific traffic condition of the main road, side road, or pedestrian crossing.

To represent these six states, a 3-bit binary encoding scheme is used. Three bits are sufficient to uniquely encode all FSM states, ensuring clear and unambiguous state transitions while keeping the hardware implementation simple and efficient.

In addition to state encoding, the traffic light outputs for both the main road and the side road are also represented using a 3-bit encoding format. This encoding is used to indicate the RED, YELLOW, and GREEN light signals. Using the same 3-bit structure for state representation and output signals improves design consistency and makes simulation results and waveform analysis easier to interpret.

Timing Behavior

The operation of the traffic light controller is controlled by a timer that determines how long the system stays in each state. A counter inside the FSM counts the number of clock cycles spent in the current state and decides when to move to the next state.

The main road green state (MG) remains active for at least 60 clock cycles. During this time, the controller keeps the main road green unless a vehicle is detected on the side road or a pedestrian presses the button. The side road green state (SG) and the pedestrian walk state (PG) each last for exactly 40 clock cycles before the controller transitions to the next state. Yellow and all-red states are used between green phases to ensure safe switching between traffic directions. These states are executed automatically as part of the FSM sequence and do not require additional timing control.

The timer is reset whenever the FSM changes state, ensuring accurate timing for each traffic phase. This timing approach guarantees correct operation of the controller and full compliance with the system timing requirements.

Results

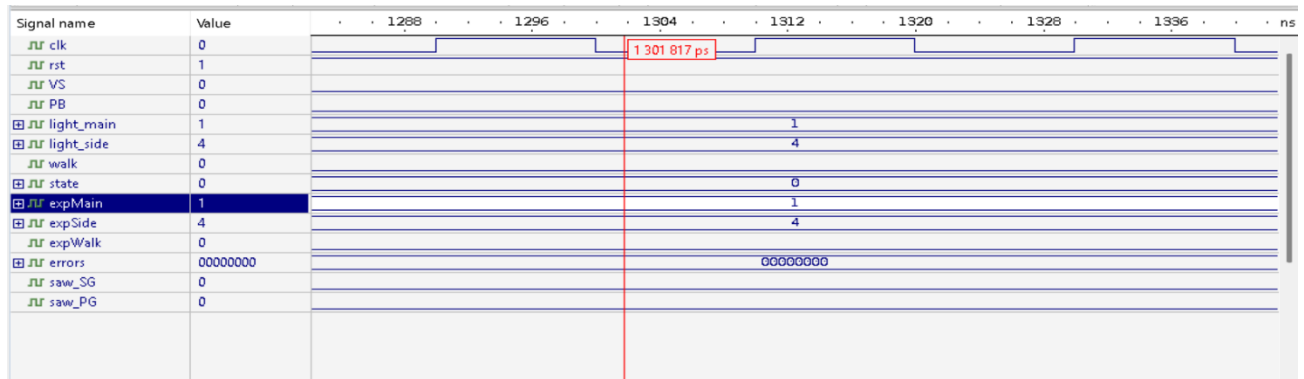


Figure 2: Default (VS = 0, PB = 0)

The waveform shows the default behavior of the traffic light controller when no side-road vehicle or pedestrian requests are present (VS = 0, PB = 0). After reset, the system remains in the Main Green (MG) state. The main road light is green, the side road light is red, and the pedestrian walk signal is inactive. The expected outputs match the actual outputs, and no errors are detected, confirming correct operation in the no-request case.

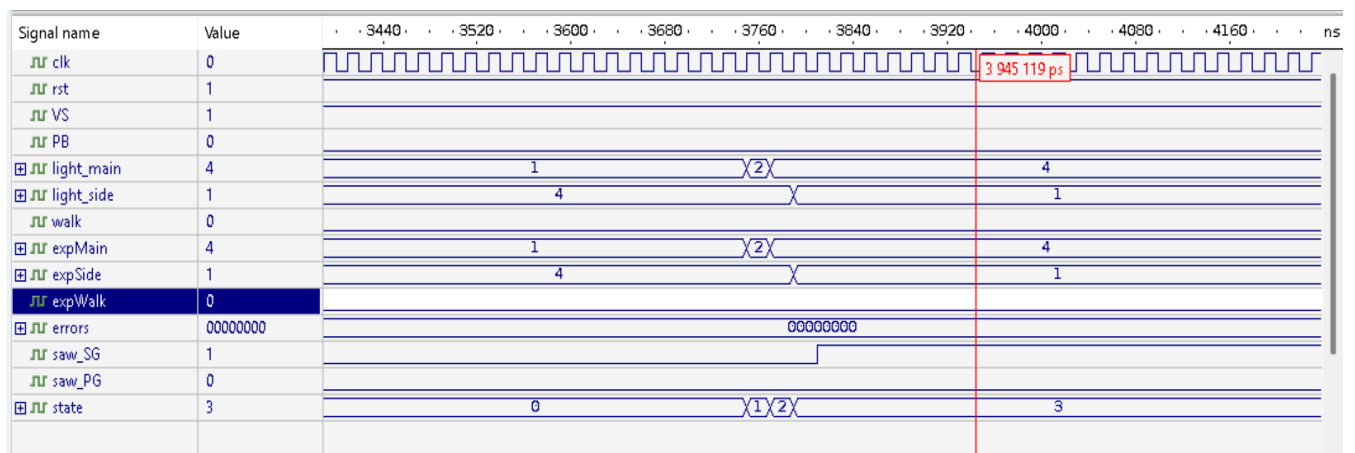


Figure 3 :Side-Road Vehicle Request (VS = 1, PB = 0)

In this case, the side-road vehicle sensor is activated while no pedestrian request is present. As shown in the waveform, the controller initially operates in the Main Green (MG) state. After the minimum green time expires, the system transitions to Main Yellow (MY), followed by the All-Red (AR) state to ensure a safe change. The controller then enters the Side Green (SG) state, where the side road traffic light turns green (light_side = GREEN), the main road traffic light turns red (light_main = RED), and the pedestrian walk signal remains inactive (walk = 0). The expected outputs match the actual outputs, and no errors are detected, confirming correct operation in the Side-Road Vehicle Request case.

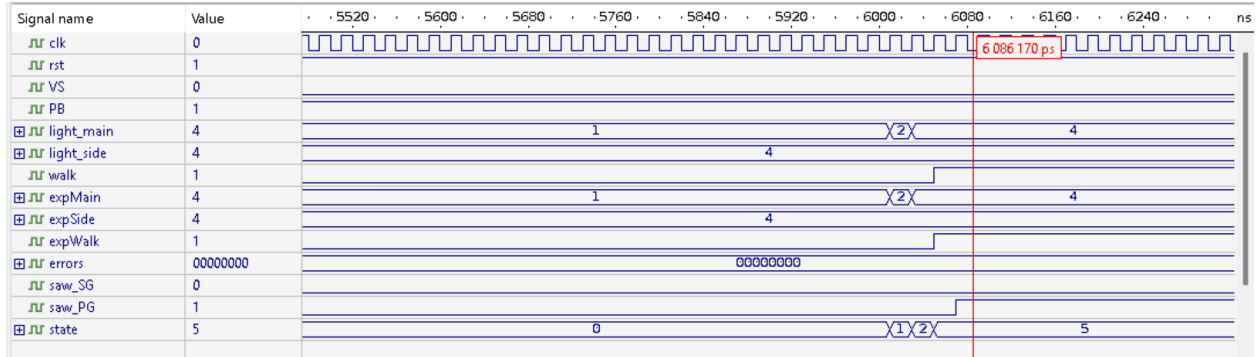


Figure 4: Pedestrian Request (VS = 0, PB = 1)

In this case, the pedestrian push button is activated while no side-road vehicle request is present. As shown in the waveform, the controller initially operates in the Main Green (MG) state. After the minimum green time expires, the system transitions to Main Yellow (MY), followed by the All-Red (AR) state. The controller then enters the Pedestrian Green (PG) state, where both traffic lights remain red and the pedestrian walk signal becomes active (walk = 1). The expected outputs match the actual outputs, and no errors are detected, confirming correct operation for a pedestrian request.

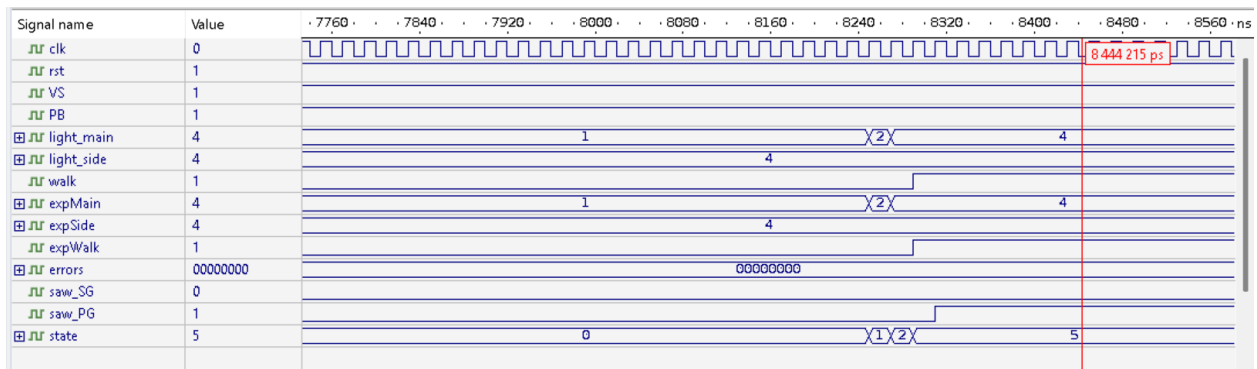


Figure 5: Request (PB = 1, VS = 1)

When both the side-road vehicle sensor (VS) and the pedestrian push button (PB) are asserted simultaneously, this test case is used to determine which request the traffic light controller responds to first. As shown in the waveform, the controller initially operates in the Main Green (MG) state. After the minimum main green duration expires, the system transitions through the Main Yellow (MY) and All-Red (AR) states before entering the Pedestrian Green (PG) state. This behavior clearly demonstrates that the controller gives priority to the pedestrian request over the side-road vehicle request. During the Pedestrian Green state, both traffic lights remain red and the pedestrian walk signal is activated, ensuring a safe pedestrian crossing. The expected outputs match the actual outputs, and no errors are detected, confirming correct priority handling.

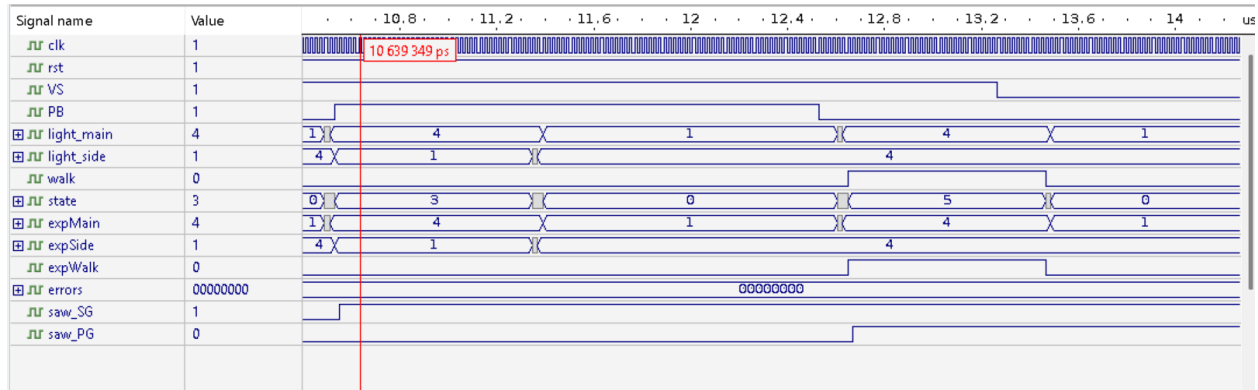


Figure 6: Pedestrian Request During Side-Green (Queued Request).

In this case, a pedestrian request ($PB = 1$) is asserted while the controller is operating in the Side Green (SG) state. As illustrated in the waveform, the pedestrian walk signal remains LOW during the SG state, indicating that the request is not served immediately. Instead, the request is stored internally as a pending (queued) request. After the Side Green duration expires and the controller transitions through the intermediate states, the queued pedestrian request is finally served by entering the Pedestrian Green (PG) state, where the walk signal becomes active. This behavior confirms the correct handling of pedestrian requests during side-road operation.

Note:

The small boxed segments that appear in the waveform represent the transition states of the FSM, specifically the Main Yellow (MY), Side Yellow (SY), and All-Red (AR) states. These short intervals are intentionally included to ensure safe switching between traffic directions. They are shown clearly in the zoomed-in waveform to demonstrate that the controller correctly responds first to the side-road vehicle request and then to the pedestrian request, following the defined state transition sequence.

Conclusion

This project presents the design and implementation of a traffic light controller with pedestrian support based on a finite state machine (FSM). The controller is responsible for managing traffic flow at an intersection that includes a main road, a side road, and a pedestrian crossing, where the main road is given priority and remains green by default.

The controller operates through several defined states, each representing a specific traffic light condition with fixed timing behavior. Different scenarios, such as side-road vehicle requests, pedestrian requests, simultaneous requests, and queued pedestrian requests, were tested and analyzed. These cases were verified using a simulation-based testbench to ensure that each state transition occurs at the correct time and follows the specified timing constraints.

Simulation results confirmed that the controller responds properly to all tested conditions, follows the required timing behavior, and transitions between states as expected. Overall, the project successfully meets the design requirements and demonstrates correct and reliable operation of the traffic light control system.